

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – OPEN DATA ANALYSIS PROJECT

Course Code:	DSA05	Course Title:	Query Processing for Data Science in Real Time Applications
CO Assessed:	CO4	Assessment:	Tool 3 – Open Data Analysis Project
Weightage:		Total Marks:	50 Marks
CO4 Statement: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.. (BL4, BL5 and BL6)			
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Section / Batch:	Slot-A 2024	Date of Submission:	21-08-2026
Faculty In-charge:	K.Veena devi	Assessment Mode:	Individual Submission

Code of Conduct

I __**P.Sunayana(192424303)**__ (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: __P.Sunayana__

Data Collection & Understanding (10 Marks)	Data Cleaning & Preparation (10 Marks)	Exploratory Data Analysis (10 Marks)	Visualization & Dashboard Design (10 Marks)	Demonstration (10 Marks)	Total (50 Marks)

2. Air Quality and Environmental Impact Assessment

Obtain air quality datasets from World Air Quality Index Project or government open-data portals and analyze pollution levels across cities or regions. Perform exploratory data analysis to identify seasonal variations, pollutant concentration patterns, and correlations between pollutants and meteorological variables. Detect anomalies, visualize geographic distributions using maps, and propose data-driven recommendations for environmental management and public health awareness.

Air Quality and Environmental Impact Assessment

1. Introduction

Air pollution is a major environmental and public health concern. An Air Quality and Environmental Impact Assessment analyzes pollution data collected from different cities or regions to understand the concentration of harmful pollutants and their relationship with weather conditions.

The dataset can be obtained from sources such as the World Air Quality Index (WAQI) Project, government open-data portals, or environmental monitoring agencies.

The main pollutants that can be analyzed include:

- PM2.5
- PM10
- NO₂
- SO₂
- CO
- O₃

Meteorological variables may include:

- Temperature
- Humidity
- Wind speed
- Rainfall
- Atmospheric pressure

2. Objectives

The main objectives of this assessment are:

1. To collect air quality data from different cities or regions.
2. To perform data cleaning and exploratory data analysis.
3. To analyze seasonal variations in pollution levels.
4. To identify patterns in pollutant concentrations.
5. To study correlations between pollutants and meteorological variables.
6. To detect unusual or abnormal pollution events.
7. To visualize air quality geographically using maps.
8. To provide data-driven recommendations for environmental management and public health awareness.

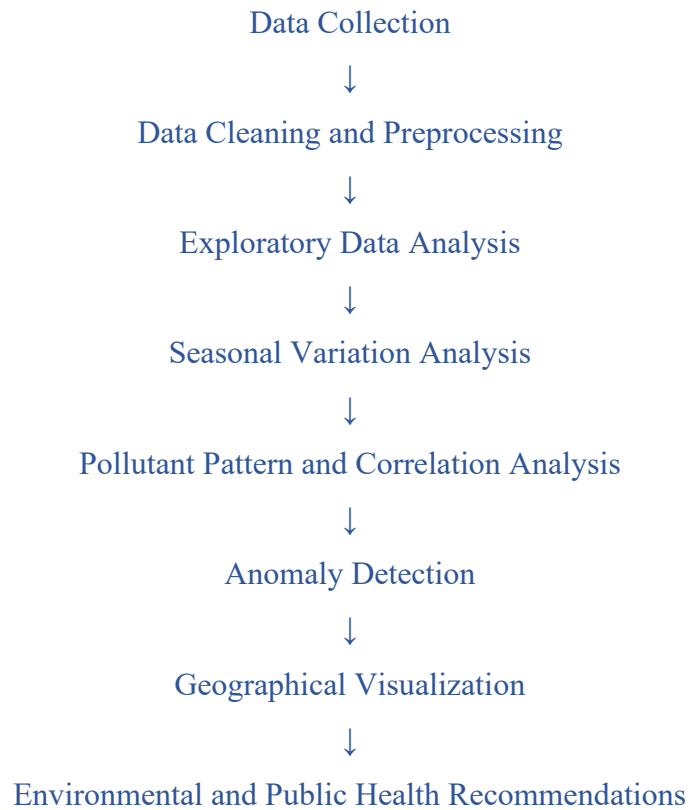
3. Dataset Description

A typical air quality dataset may contain the following attributes:

Attribute	Description
Date	Date of air quality measurement
City/Region	Location of monitoring station
PM2.5	Fine particulate matter concentration
PM10	Coarse particulate matter concentration
NO ₂	Nitrogen dioxide concentration
SO ₂	Sulfur dioxide concentration
CO	Carbon monoxide concentration
Temperature	Atmospheric temperature
Humidity	Percentage of moisture in air
Wind Speed	Speed of wind
Latitude	Geographic latitude
Longitude	Geographic longitude

4. Methodology / Workflow

Overall Workflow



5. Data Cleaning and Preprocessing

Before analysis, the dataset should be cleaned.

Step 1: Check Missing Values

Identify missing values in pollutant and weather columns.

Possible methods:

- Remove records with excessive missing values.
- Replace missing numerical values using mean or median.
- Use interpolation for time-series data.

Step 2: Remove Duplicate Records

Duplicate observations can affect average pollution calculations and must be identified and removed.

Step 3: Standardize Date Format

Convert all date values into a standard date format.

For example:

2026-01-15

This helps in monthly, seasonal, and yearly analysis.

Step 4: Check Data Types

Ensure that:

- Date is stored as a date/time variable.
- Pollutant concentrations are numerical.
- Latitude and longitude are numerical.

Step 5: Detect Invalid Values

Check for unrealistic values such as:

- Negative PM2.5 concentration
- Missing coordinates
- Extremely high or low measurements

6. Exploratory Data Analysis

Exploratory Data Analysis (EDA) helps understand the overall distribution and behavior of air pollutants.

Important analysis includes:

- Mean
- Median
- Minimum
- Maximum
- Standard deviation
- Distribution of pollutant values

Recommended Visualizations

- Histogram
- Box plot
- Line chart

- Bar chart
- Density plot

For example, a histogram of PM_{2.5} can show whether pollution values are normally distributed or highly skewed.

A box plot can identify extreme pollution measurements.

7. Seasonal Variation Analysis

Air pollution often changes according to seasons.

The data can be grouped into seasons such as:

- Summer
- Monsoon
- Winter
- Post-monsoon

Calculate the average concentration of pollutants for each season.

Example

Season	Average PM _{2.5}
Summer	55
Monsoon	30
Winter	120
Post-monsoon	80

A line chart or grouped bar chart can be used to compare seasonal pollution levels.

Interpretation

Higher pollution levels during winter may occur because of:

- Low wind speed
- Temperature inversion
- Vehicle emissions
- Industrial emissions

- Reduced dispersion of pollutants

Lower pollution levels during rainy seasons may occur because rainfall can help remove particulate matter from the atmosphere.

8. Pollutant Concentration Pattern Analysis

The concentration of different pollutants can be compared across cities or regions.

For example:

City	PM2.5	PM10	NO ₂	SO ₂
City A	85	150	42	15
City B	45	90	30	10
City C	120	200	65	25

Recommended visualizations:

- Grouped bar chart
- Box plot
- Heatmap
- Line chart

This analysis helps identify which city or region experiences the highest pollution.

9. Correlation Between Pollutants and Meteorological Variables

Correlation analysis can identify relationships between air pollutants and weather conditions.

For example:

- PM2.5 vs Temperature
- PM2.5 vs Humidity
- PM2.5 vs Wind Speed
- NO₂ vs Temperature
- PM10 vs Rainfall

A correlation matrix and heatmap are useful for this analysis.

The correlation coefficient ranges from:

$$-1 \leq r \leq +1$$

Interpretation

Correlation Value	Interpretation
Close to +1	Strong positive relationship
Close to 0	Weak or no relationship
Close to -1	Strong negative relationship

Example

A negative correlation between PM_{2.5} and wind speed may indicate that stronger winds help disperse pollutants.

A positive correlation between PM_{2.5} and PM₁₀ may indicate that both pollutants are influenced by similar pollution sources.

10. Anomaly Detection

Anomalies are unusual pollution measurements that are significantly different from normal values.

For example:

- Normal PM_{2.5}: 40–100
- One observation: 500

The value of 500 may be considered an anomaly and should be investigated.

Methods for Detecting Anomalies

1. IQR Method

$$IQR = Q3 - Q1$$

Upper boundary:

$$Q3 + 1.5 \times IQR$$

Values above the upper boundary can be treated as potential outliers.

2. Z-Score Method

$$Z = \frac{X - \mu}{\sigma}$$

Values with:

$|Z| > 3$

may be considered unusual observations.

3. Time-Series Analysis

A sudden increase in pollution compared with previous days or months can indicate an unusual event.

Possible causes include:

- Industrial accidents
- Wildfires
- Construction activities
- Festivals involving fireworks
- Traffic congestion
- Dust storms

11. Geographic Distribution Using Maps

Since air quality data contains location information, geographical visualization is very useful.

Required columns:

- City
- Latitude
- Longitude
- Pollutant concentration

Recommended Map Types

1. Bubble Map

The size of each bubble represents the pollution level.

Larger bubbles indicate higher pollutant concentrations.

2. Heat Map

A heat map identifies pollution hotspots.

High-concentration areas can be easily identified.

3. Choropleth Map

Different regions can be shaded according to their average pollution levels.

Benefits

Maps help decision-makers:

- Identify pollution hotspots.
- Compare cities and regions.
- Locate areas requiring immediate action.
- Plan monitoring stations.

12. Data-Driven Recommendations

Based on the analysis, the following recommendations can be proposed.

1. Strengthen Monitoring in High-Pollution Areas

Cities or regions with consistently high PM_{2.5} or PM₁₀ levels should receive additional monitoring stations.

2. Control Vehicle Emissions

If NO₂ and particulate matter are high in traffic-heavy areas, authorities can:

- Promote public transportation.
- Encourage electric vehicles.
- Improve traffic management.
- Implement vehicle emission checks.

3. Regulate Industrial Emissions

Regions with high SO₂ or particulate matter may require:

- Regular industrial inspections.
- Emission control technologies.
- Stricter pollution standards.

4. Seasonal Pollution Management

If pollution increases during particular seasons, preventive measures should be implemented before the high-pollution period.

For example:

- Restrict open burning.
- Control construction dust.

- Increase public transport services.
- Monitor industrial activities.

5. Public Health Awareness

During periods of high pollution, the public can be informed about:

- Air quality levels.
- Reducing outdoor activities when pollution is severe.
- Protecting vulnerable groups such as children and elderly people.
- Using appropriate preventive measures when needed.

13. Expected Visualizations

The following visualizations can be included in the assessment:

Analysis	Visualization
Monthly pollution trend	Line Chart
Distribution of PM2.5	Histogram
Outlier detection	Box Plot
Seasonal comparison	Bar Chart
Pollutant relationships	Correlation Heatmap
Pollutant vs weather variables	Scatter Plot
Pollution across cities	Grouped Bar Chart
Geographic pollution distribution	Bubble Map / Heat Map
Pollution hotspots	Geographic Heat Map

14. Expected Outcome

The project will provide meaningful insights into:

- Cities or regions with high pollution levels.
- Pollutants that contribute most to poor air quality.
- Seasonal variations in air pollution.

- Relationships between weather conditions and pollution.
- Unusual pollution events or anomalies.
- Geographic pollution hotspots.

Conclusion

This Air Quality and Environmental Impact Assessment uses data cleaning, exploratory data analysis, grouping, correlation detection, anomaly detection, and geographical visualization to study air pollution patterns. The results can help environmental authorities identify high-risk regions, understand seasonal pollution behavior, and implement effective pollution-control strategies. It also supports public health awareness by providing clear information about periods and locations with poor air quality.